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April 3, 2025

Refining the Effective Radius Measure for a Sample of Early-Type Galaxies Through
the Use of GALFIT

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Abstract

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The co-evolution of supermassive black holes (SMBH) and their host galaxies remains an active area of research. In this paper we seek to improve the accuracy of the galaxy bulge radius measurements, by applying multi-component fits to a sub-sample of 8 galaxies from the *Atlas^{3D}* survey (Cappellari et al. (2011)), using GALFIT, a 2D luminosity function fitting software (Peng et al. (2010)). After fitting these 8 galaxies, and by including galaxies from Kesha Patel's thesis, we were able to determine that on average, galaxies modeled through GALFIT had an effective radius (R_e) 3.7 ± 0.7 times smaller than those obtained from fitting single Gaussian function fits.

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Chapter 1

Introduction

It has been well-established that there is a supermassive black hole (SMBH) at the center of every galaxy, and that the evolution of the SMBH and its host galaxy are fundamentally intertwined (Ferrarese & Merritt (2000) and Gebhardt et al. (2000)). SMBHs are characterized as black holes with a mass greater than $10^5 M_{\odot}$. SMBHs can be characterized as either active or quiescent. Those that are active are actively accreting mass are known as Active Galactic Nuclei (AGN) (Keel (1996) and Rees (1984)). As matter falls towards the SMBH in an AGN, it forms a disk-like structure (an accretion disk) where matter orbits the SMBH until it loses enough energy to fall into the SMBH, causing its mass to increase (Ferrarese & Ford (2005)). As these accretion disks orbit the SMBH, they are heated to extreme temperatures causing them to glow as bright as the entire host galaxy (Wu et al. (2015)). However, most galaxies in the local universe don't contain an AGN (Paspaliaris et al. (2023)); galaxies with a black hole that is not actively accreting material are known as quiescent galaxies (Jr et al. (2017)).

Due to how compact SMBHs are, regardless of their large mass, their sphere of influence (SOI), which is the region gravitationally affected by the SMBH, is extremely small when compared to the size of the galaxy, making it nowhere near

large enough to have influence over the whole galaxy (Graham et al. (2016)). This lack of gravitational connection between the SMBH and the galaxy at large scale, indicates that the coevolution of the SMBH and the galaxy is not controlled by gravity, but by some other physical mechanism that we do not yet fully understand (Kormendy & Ho (2013)). One method of probing this co-evolution is through investigation of correlations that have been found to exist between the mass of the SMBH (M_{BH}) and properties of the host galaxy, such as: bulge stellar velocity dispersion (σ_*) (Ferrarese & Merritt (2000)); total bulge mass (Kormendy & Richstone (1995)); and bulge luminosity (Dressler (1989)). The existence of these relations implies that the SMBH and their host galaxy likely evolve together, and therefore understanding these relations is essential to understanding SMBH and galaxy evolution.

Of particular interest in this work is the $M_{BH} - \sigma_*$ relation, which compares the mass of the SMBH to the stellar velocity dispersion in the bulge of the host galaxy, σ_* , which is simply the average relative motion of all the stars in the central region of the galaxy. The reason this relation is so important is that, although all scaling relations point to a co-evolution between the SMBH and the galaxy itself, the $M_{BH} - \sigma_*$ relation is thought to provide the tightest relation (Ferrarese & Merritt (2000), Gebhardt et al. (2000), and Beifiori et al. (2012)). Figure 2.3 shows the fit to the $M_{BH} - \sigma_*$ relation from Batiste et al. (2017a). As can be seen in the plot, the correlation is clear, but the scatter around the best fit is still far too large to properly constrain this relation (Pacucci et al. (2018) and Batiste et al. (2017b)), which is necessary in furthering our understanding of the underlying physics. This scatter originates from both observation errors, such as the uncertainties in measures of the M_{BH} and σ_* (Gebhardt et al. (2000)), but there also seems to be an intrinsic scatter than cannot be accounted for simply by errors in measurements. By reducing measurement error, we leave only the intrinsic scatter in the relation, which would allow the physical processes explaining this intrinsic scatter to be more effectively investigated.

It is normal to think of galaxies as just a gravitationally bound collection of stars floating around the universe, but in reality they are a lot more complicated than that. Galaxies are split into several different structural components, each with different properties. A great example of the complexity of galaxies is our very own, the Milky Way (MW) [Figure 3.3]. Our solar system resides in the disk of the MW, which is a flat rotating region where most of the new stars are created. As part of the disk we have spiral arms, which are denser regions where the star formation rate is even higher than in other regions. As gas enters these arms it is compacted which leads to the birth of new stars. Towards the middle of the galaxy we have the bulge of the galaxy, which hosts the SMBH, in our case Sagittarius A*. The bulge is a spherical and much denser region than the rest of the galaxy which mostly contains older stars as there is little to no star formation actually happening in the bulge. Connecting the bulge and the spiral arms of our galaxy we have an extensive bar, which is a long stretched-out structure thought to funnel gas from the disk of the galaxy into the bulge (Rodriguez-Fernandez & Combes (2008)) (see Section 3.1 for further descriptions of galaxy structure). What's most important for our project is the bulge of the galaxy. The MW is known as a disk galaxy, as its structure is dominated by the extended disk, but other galaxies are dominated by the bulge with little to no rotating disk. These galaxies are known as ellipticals. The velocity dispersion (σ_*) described in this work specifically refers to the average velocity dispersion in this spheroid region.

The first part of the $M_{BH} - \sigma_*$ relation involves accurate measurement of the M_{BH} . This is either done by stellar dynamical modeling (SDM) or reverberation mapping (RM). SDM models the way stars that are gravitationally affected by the SMBH move under its influence and compare it to simulations of the way stars would move around SMBH with different masses to determine the best fitting M_{BH} (Siopis et al. (2009)). This method only works for the most nearby galaxies as it requires you to be able to resolve the sphere of influence (SOI) of the SMBH, which is incredibly small due

to how compact SMBHs are. The second method, RM, can only be performed on AGN-host galaxies as it requires the light produced from the accretion disk, which is known to randomly vary in intensity, as well as the “echoed” light from the fast moving gas around it, known as the broad line region (BLR). This “echo” is caused by emitted light in the BLR after the gas in the region gets excited by the light from the AGN accretion disk. (see section 2.3 for further discussion of mass measurement). Both mass measurement methods directly probe the gravitational SOI of the SMBH, and a great deal of work has been done to improve the accuracy of these methods, to the point where they are no longer the dominant source of measurement error in the $M_{BH} - \sigma_*$ relation.

Since the M_{BH} has been sufficiently well constrained, to further constrain the $M_{BH} - \sigma_*$ relation we need to reduce error in σ_* , which requires an accurate measure of the radius of the bulge of the galaxy. The radius of the bulge is defined as the distance from the center that contains half the total light produced by the bulge, and is called the effective radius (R_e). The R_e can be measured by fitting the light distribution in galaxies so as to separate it into its different structural components. Accurately fitting the light distribution of galaxies is essential to measuring accurate R_e , which would then allow σ_* to be further constrained (see sections 3.2 and 3.3 for further detail).

Prior work by Batista et al. (2017b) to improve measurements of σ_* , in order to recalculate the $M_{BH} - \sigma_*$ relation, focused on a sample of 10 AGN-host galaxies, all of which had well constrained M_{BH} measures obtained through RM. The AGN-host galaxies had accurate R_e measures done by Bentz et al. (2009), which involved multi-component light profile fitting for each galaxy using a software called GALFIT (Peng et al. (2010)). GALFIT allows one to decompose each galaxy’s surface brightness distributions into multiple structural components. Each of these components is modeled by a different function. This method allows you to account for the complex

structure of galaxies, such as bulges, disks, bars, and others, a lot more accurately than a single component fit (Batiste et al. (2017b)). These accurate R_e measures allowed for the $M_{BH} - \sigma_*$ relation for AGNs to be more accurately constrained. Batiste et al. (2017b) compared their relation for the AGN sample to the one for the quiescent sample, which is necessary for the calibration of the M_{BH} through RM (see section 2.3.2 for further explanation), but found a systematic bias in the sample. The bias originated in the quiescent sample, which did not have measurements of R_e determined through this method, and so the values were far more poorly constrained.

The sample of quiescent galaxies examined by Batiste et al. (2017a) was obtained from the *ATLAS*^{3D} survey. (Cappellari et al. (2011)) They sought to obtain spatially resolved spectroscopy for a sample of nearby quiescent galaxies. Their goal was to study the kinematics of these galaxies and not to model the structural components of these galaxies, so even though they present R_e values for all of these galaxies, their determination was deliberately simplified. To determine the R_e they chose to fit single a gaussian function to each galaxy and treated the radius of that component to be the R_e of the bulge (see section 4.1 for further explanation). This led to inaccurate R_e measurements, as was demonstrated by Batiste et al. (2017b) who showed that a single component fit which was done to NGC 5373 by Cappellari et al. (2011) led to an over estimation of the R_e by 5 times when compared to a 2 component fit done by Bentz et al. (2014). So, as the AGN sample has already been accurately evaluated, further work on constraining the $M_{BH} - \sigma_*$ relation needs to focus on the quiescent sample of galaxies.

In this project, we investigate how improving the fits of a sub-sample of 8 galaxies, selected from the *Atlas*^{3D} sample, will affect the measured R_e . To do this, we generate 2 component fits for Hubble Space Telescope images of each galaxy in this sub-sample using GALFIT (Peng et al. (2010)). The goal is not to perfect the fits for these 8 galaxies, but rather to evaluate the statistical impact of adding at least 1 more

components to each fit compared to doing single gaussian function fits. The addition of one more component can allow a more classical bulge, if present, to be fit independently of the disk of the galaxy which would lead to significantly more accurate measures of the R_e needed for obtaining σ_* . As a result, many of the fits presented in this study will not fully model the galaxies, but they will provide insight into whether more accurate modeling across the full quiescent sample is warranted, or if the simpler single-component fits are sufficient.

This paper is divided into 7 different chapters. The second chapter describes galaxy morphology, the $M_{BH} - \sigma_*$ relation, and dives deeper into the methods used to measure SMBH mass. This is then followed by a description of luminosity profile fitting, as well as an explanation of the software GALFIT in chapter 3. The fourth chapter then covers the *Atlas*^{3D} survey, the Hubble Space Telescope, and contains descriptions for each galaxy used in this project. The fifth chapter walks through the fits for each galaxy, and the 6th is an analysis of these results. Finally the 7th chapter is a conclusion with an overview of this project, as well as future work.

Chapter 2

Galaxy Structure and Black Hole Scaling Relations

2.1 Galaxy Morphology

Galaxy morphology refers to the study of the structure and appearance of galaxies. This study originated in the 1920s when Edwin Hubble noticed that galaxies had a variety of different shapes, and decided to organize and classify them (Hubble (1936)). This was done through the Hubble tuning fork [Figure 2.1], a diagram which he originally believed to be an evolutionary sequence for galaxies, where they began as elliptical (E) galaxies, then evolved into spiral (S) galaxies, passing through a lenticular (S0) phase as a transition point between the two. We now understand that the evolution of galaxies is more complicated, but the diagram is still widely used to classify galaxies.

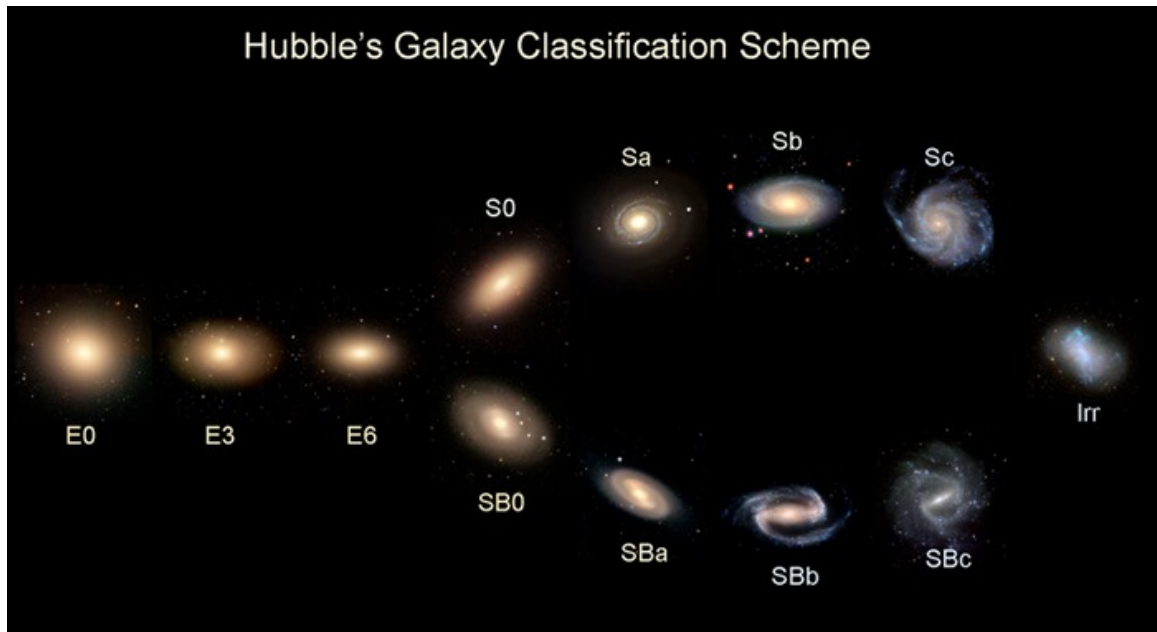


Figure 2.1: Hubble tuning fork for galaxy classification¹

Galaxies are split into 4 main categories: elliptical (E), lenticular (S0), spiral (S), and irregular (I). Elliptical galaxies are generally simple spheroids, supported by gas pressure, with minimal rotation, but they can also be made up of more complicated structure, containing, bulges, disks, and others. Spiral galaxies are more complicated, having at least a bulge, a disk, and spiral arms. The bulge is the central region of the galaxy which is a spheroid essentially identical in structure to an elliptical galaxy. This region is more dense than the rest and is classified as having random stellar motion. The disk of the galaxy is a prominent, flat, rotating component that contains most of the gas in these galaxy and extends past the bulge. The spiral arms are regions of high density within the disk, which lead to higher rates of active star formation, and give spiral galaxies their characteristic appearance (Graham (2013)). These galaxies however, can contain many other structural components such as pseudo-bulges, bars, and rings (Erwin & Sparke (1999)). Pseudo-bulges are central structures

¹<https://itu.physics.uiowa.edu/labs/advanced/classifying-galaxies/part-1-hubbles-tuning-fork>

that appear to be similar to classical bulges, but also display disk-like properties. Bars are elongated features that stretch across the centers of disks and can play major roles in the dynamics of the gas in a galaxy. Finally rings are circular features that appear in galaxies with disks, they are a larger concentration of stars and gas, which are often sites of enhanced star formation (Fernandez et al. (2021)).

Lenticular galaxies, also known as S0, are considered to be a transition state between spiral and elliptical galaxies (Hubble (1936)). Lenticular galaxies are characterized by having a bulge and a disk similar to spiral galaxies, but they lack spiral arms, and have older stellar populations similar to elliptical galaxies. This older star population comes from a lack of gas which is a common characteristic of both elliptical and lenticular galaxies. This leads to little to no new stars forming which results in all the stars in the galaxy being on average older (DeGraaff et al. (2007)).

The final type of galaxies are irregulars, which essentially encompasses anything that doesn't fit into the other categories. These galaxies are structurally complex as they have not yet virialized. What is meant by a galaxy being virialized is that it has reached a kind of gravitational equilibrium where the internal motions and gravitational forces are balanced. Irregular galaxies are believed to have gone through a recent gravitational interaction and have not yet had enough time to return to this state of equilibrium which gives them their irregular shape (Morgan & Mayall (1957)).

In the Hubble classification system galaxies are also broadly divided into early-type and late-type based on when Hubble believed they existed in the evolutionary sequence of a galaxy. Early-type galaxies include both elliptical and lenticular galaxies and they are dominated by older star populations and contain little to no ongoing star formation. Late-type galaxies, on the other hand, include spiral and irregular galaxies. These are disk dominated galaxies and display prominent spiral arms, dust lanes, and active star formation (Baldry (2008)). This evolutionary sequence is now known to not be real, but regardless the naming convention persists.

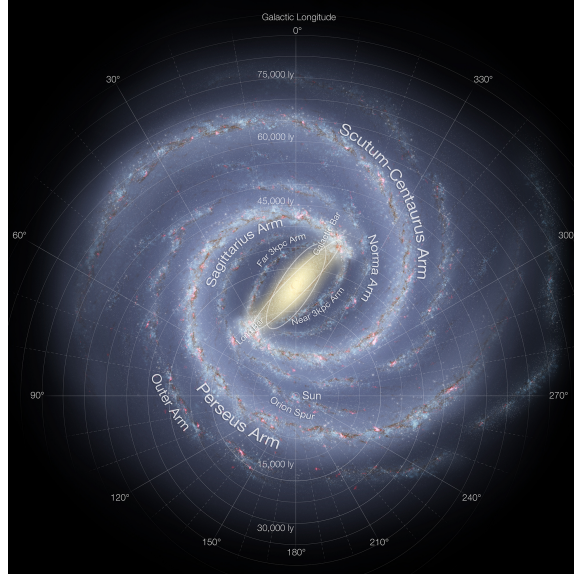


Figure 2.2: Artist depiction of the Milky Way (MW)²

2.2 The $M_{BH} - \sigma_*$ relation

The correlation between the mass of the SMBH (M_{BH}) and the stellar velocity dispersion of a galaxy's bulge (σ_*) continues to be thought of as one of the most robust indicators for the co-evolution of galaxies and their central SMBH (Ferrarese & Merritt (2000) and Gebhardt et al. (2000)). Accurate calibration of this relation is essential for a better understanding of the physical processes that underlie this co-evolution. Batiste et al. (2017b) worked to recalibrate the $M_{BH} - \sigma_*$ relation by investigating a sample of 16 reverberation-mapped AGNs and 32 quiescent galaxies. The main finding of their project was that the slope of the AGN $M_{BH} - \sigma_*$ relation was consistently shallower than that for quiescent galaxies [Figure 2.3]. They believed this to be caused by systematic inconsistencies in measures of the R_e in the quiescent sample of galaxies. Further constraining the $M_{BH} - \sigma_*$ relation for both AGN and quiescent galaxies would require a sample of galaxies with σ_* obtained from integral field spectroscopy (IFS), which can create spatially resolved spectra of the entire galaxy, and well constrained R_e measures.

²<https://webbtelescope.org/contents/media/images/4186-Image>

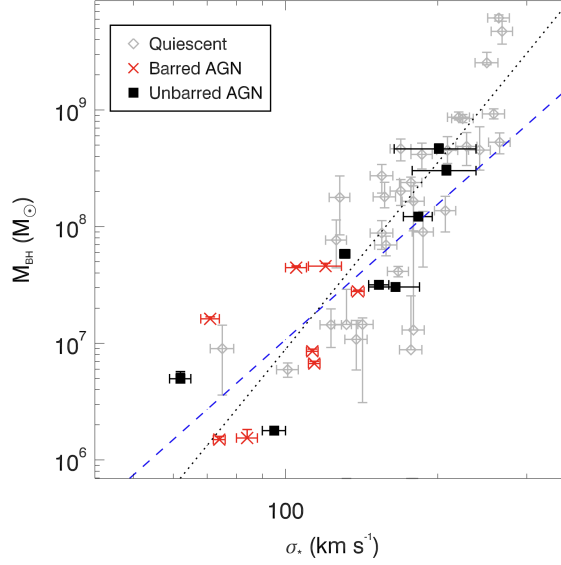


Figure 2.3: $M_{BH}-\sigma_*$ relation for quiescent galaxies (gray) and AGN (red for unbarred and black for barred) plot from Batiste et al. (2017b). The adopted best fit for the quiescent sample is shown as the dotted line and for the AGN sample is the dashed line.

2.3 Measuring SMBH Mass

2.3.1 Stellar Dynamical Modeling (SDM)

SDM involves mapping the average motion of stars within the SMBH's region of influence and modeling their orbits around SMBHs of different masses to identify the best match (Siopis et al. (2009)). Due to how compact SMBHs are, their SOI is extremely small when compared to the rest of the galaxy. This size makes resolving the SOI for any but the most nearby galaxies impossible with current telescopes, making it only feasible for nearby galaxies. AGNs are most commonly found outside of the local universe (Paspaliaris et al. (2023)), but even for the rare AGN that is close enough to spatially resolve the SMBH SOI, the light emitted by the accretion disk can completely mask the SOI making it difficult to observe the host galaxy in that region. This makes SDM generally feasible only for quiescent galaxies in the very local universe.

2.3.2 Reverberation Mapping (RM)

RM is a method of measuring the mass of AGN whose spectra exhibit broadened emission lines due to Doppler broadening from gas orbiting rapidly under the gravitational influence of the SMBH. The flux from AGN accretion disks has been found to vary randomly (King et al. (2004)), and this variation is then “echoed” in the emission lines from this rapidly orbiting gas, known as the broad line region (BLR). RM directly probes the gravitational potential of the SMBH by using the time-lag between the changes in flux of the SMBH accretion disk and the corresponding change in flux observed from BLR emission lines. Under the assumption that light travels at the speed of light between the accretion disk and the BLR, the timing differential can be used to determine the size of the BLR (R_{BLR}). Via the virial theorem, we can use this connection to determine M_{BH} .

$$M_{BH} = f \left[\frac{\Delta V^2 R_{BLR}}{G} \right] \quad (2.1)$$

ΔV is the line width of the Doppler-broadened emissions from the BLR which tells us how fast the gas is moving, G is the gravitational constant, the radius of the BLR (R_{BLR}) can be determined by the time lag in flux changes, and f is a multiplicative scale factor that accounts for the geometry and kinematics of the BLR. From RM we can accurately determine the so-called virial product given by M_{BH}/f and shown in square brackets in eq. 21. However accurate determination of M_{BH} depends on f . Little is known about the geometry of the BLR, and it cannot yet be directly observed, so instead f is estimated as the multiplicative factor needed to bring the $M_{BH} - \sigma_*$ relation for AGN and quiescent galaxies into agreement (Onken et al. (2004)). This is done under the assumption that the $M_{BH} - \sigma_*$ relation should be the same for both quiescent galaxies and those hosting AGN, though this assumption continues to be investigated (Woo et al. (2013)). The exact form of the $M_{BH} - \sigma_*$ relation can lead

to significant changes in f , and any measurement errors can mask differences between AGN and quiescent $M_{BH} - \sigma_*$ relations. Batista et al. (2017b) for example, showed the the $M_{BH} - \sigma_*$ relation for AGN and quiescent galaxies showed two different fits, but these fits were statistically indistinguishable. However, the systematic bias present in the quiescent sample makes matter more complicated.

Chapter 3

Modeling the Galaxy Light Distribution

3.1 Luminosity Profile Fitting

Luminosity profile fitting is a technique used to quantitatively describe how the brightness of galaxies varies with radius. By fitting observed galaxy brightness distributions with different functions one can decompose the galaxy into its base structural components. The following subsection describes the most commonly used function for these fittings, these are the functions that will be used in this work.

3.1.1 Sèrsic Profile

The Sèrsic profile will be the most commonly used function when attempting to decompose our sample of galaxies, as its vast range of input variables allows it to model a wide range of galaxy structures with high accuracy. This function follows a power law of the following form:

$$\Sigma(r) = \Sigma_e \exp \left[-\kappa \left(\left(\frac{r}{r_e} \right)^{1/n} - 1 \right) \right] \quad (3.1)$$

in this equation Σ_e is the pixel surface brightness at the effective radius (r_e), the radius at which half of the light is contained. The Sèrsic exponent (n) is often called the concentration parameter as it dictates how steep or shallow the function is. A large n corresponds to a steep inner profile with extended outer wings, whereas a small n leads to a shallow inner profile with steep truncation at large r . For $n = 4$ the Sèrsic profile becomes known as a de Vaucouleurs profile, which is often used to fit galactic bulges or simple elliptical galaxies. This profile leads to a steep central concentration of light with extended outer wings which can capture the centrally concentrated light in these structural components (de Vaucouleurs et al. (1991)). More generally, pseudo-bulges are thought to be fit by $n < 2$ and classical bulges are fit by $n > 2$ (Fisher & Drory (2010)). An index of $n < 0.5$ creates a Sèrsic profile that can be used to fit galaxy bars.

3.1.2 Exponential Function

The exponential is a special case of the Sèrsic profile with $n = 1$ (Simard et al. (2011)) and is usually used to fit the disk of a galaxy. This profile follows an equation of the form:

$$\Sigma(r) = \Sigma_0 \exp(-r/r_s) \quad (3.2)$$

the one main difference between this and simply using a Sèrsic with n fixed to 1, is the radius which in this case refers to the scale length r_s , which differs from r_e by the following relation $r_e = 1.678r_s$.

3.2 Galfit

Galfit is a two-dimensional fitting algorithm widely used for photometric decomposition of galaxy structure (Peng et al. (2010)). It allows multiple components, represented as different surface brightness profiles, to be simultaneously fit onto a galaxy image,

enabling detailed decompositions of its substructure. These different surface brightness profiles of different components are fit by specific functions detailed in the previous sections. GALFIT uses a reduced chi squared value (χ^2_ν) to determine how well the specific combination of luminosity profiles is fitting the total light distribution of the galaxy. It takes in the user input estimates for what functions should be used to fit the galaxy, as well as the parameters for each of those functions, then it adjusts the fit parameters of each function to find what set results in a χ^2_ν closest to 1. This goodness of fit measure provides an estimate of how well the components are fitting the total light distribution of the galaxy, but it can be sensitive to the presence of stars or other galaxies in the field of view, so instead of solely depending on the χ^2_ν we carefully assess the residuals from the fit to see how well the light is being fit.

The fitting process in GALFIT requires the user to upload an image of the galaxy, which in our case is a Hubble Space Telescope (HST) image downloaded from the Mikulski Archive for Space Telescopes ¹, a PSF image, which describes how the light from an unresolved point source is distributed on the CCD (further explained in section 2.3.3), and a parameter file which dictates what functions GALFIT will use in its analysis. This parameter file is the only thing you need to edit between fits, and it allows you to adjust not only what functions are used, but the values of each function. After fitting a galaxy, GALFIT outputs the original image, the model, and a residual image, which shows what light remains when the model is subtracted from the original galaxy. It also produces an output parameter file giving the best fitting values for that run. This residual image plays a critical role in not only assessing but also refining the quality of the light profile fit. The residual highlights any structure that may have not been adequately captured by the current model, as well as any structures that have been over subtracted. By looking for these structures in the residual image you are able to alter the input parameter file, by either removing or adding components,

¹<https://mast.stsci.edu/portal/Mashup/Clients/Mast/Portal.html>

so that the model becomes more physically meaningful and better corresponds to the morphology of the galaxy being fit. (Peng et al. (2010)).

3.2.1 Point Spread Functions

The point spread function, known as the PSF, is a model that needs to be inputted into GALFIT, along with the galaxy being fit, which accounts for the response of a telescope to a point source. These point sources, such as distant stars, can be spread out due to the distortions caused by the telescope itself. Point sources being so small and so distant we would expect it to simply appear as a single pixel of light, but because of a telescope's optics we see a characteristic diffraction pattern which needs to be accounted for.

When deciding on how to go about obtaining the PSF for these galaxies two different methods were considered. The first involved using GALFIT to fit a region of the image which contained a star. This star would then be fit with multiple gaussian components until it appeared to sufficiently model the distribution of light from the star (Bonfini et al. (2021)). This method would require creating a new PSF for every single galaxy, which would be problematic as some images had stars in the field that could easily be modeled while others did not. The second method was simply downloading a PSF developed by HST specifically for the instrument used to take the images. This downloaded PSF was used in the modeling of all galaxies in the sample. This PSF is widely accepted in literature, and was specifically designed to account for the complicated optic effects of the HST instruments. As can be seen in Figure 3.1 the PSF developed by HST is highly complex, and trying to replicate this by simply using Gaussians did not capture any of the complexity.

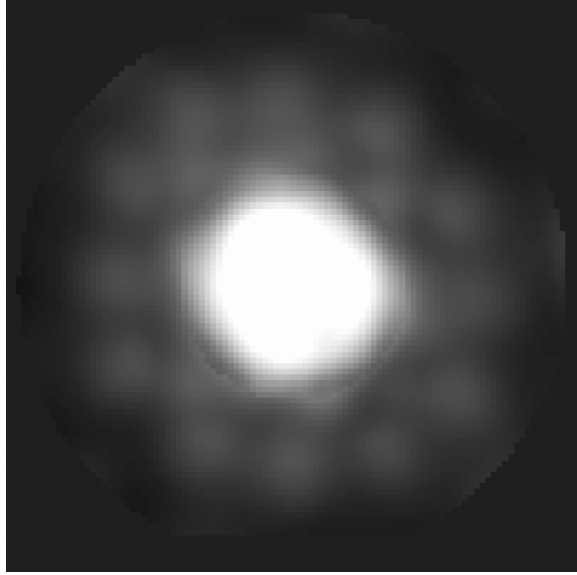


Figure 3.1: HST developed PSF for the WFC3 instrument.

Chapter 4

The Sample of Galaxies

4.1 *Atlas*^{3D}

The sample identified in this work is drawn from the *Atlas*^{3D} survey, which analyzed 260 early-type, quiescent galaxies (ETGs) located in the local universe ($distance < 42Mpc$). The primary goal of the project was to obtain spatially resolved kinematic data for each galaxy by combining high-quality integral field spectroscopy (IFS) with extensive photometric observations. The main reason for selecting my sample of galaxies from the *Atlas*^{3D} survey was that previous work done to refine the $M_{BH} - \sigma_*$ relation done by Batiste et al. (2017b) showed that there was a systematic bias present in the *Atlas*^{3D} sample, and in this experiment we seek to investigate the bias.

IFS, which can capture spatially resolved spectra over an entire galaxy, was used in both the *Atlas*^{3D} sample as well as the AGN sample explored by Batiste et al. (2017b) and Bentz et al. (2013). Although this spectral data is highly accurate, the original kinematic analysis was limited in its use for constraining σ_* due to the assumption made by the *Atlas*^{3D} collaboration, that every galaxy could be adequately modeled by a single gaussian component. As previously discussed, galaxies are significantly more complicated than this, so the R_e determined by this method are a lot less accurate

than fitting each component separately.

4.2 Hubble Space Telescope

For this analysis we used data obtained by the Hubble Space Telescope (HST). HST provides substantial advantages over ground-based telescopes due to its positioning in low Earth orbit allowing it to gather its data without dealing with the effects of looking through Earth's atmosphere. This positioning eliminates the distortion that would be caused by the Earth's atmosphere, allowing for higher definition images to be obtained across a wide range of wavelengths. Data from HST may have a lower field of view (FOV) when compared to ground-based telescopes, but the advantages of having a higher spatial resolution far outweigh any problems that may arise from the reduced FOV. Since all these images still extend far past the bulge for every galaxy in the sample, there is no real consequence to using images with a reduced FOV.

For this project we will use images taken on the WFC3 instrument which is capable of covering ultraviolet (UV), visible, and near-infrared (NIR) wavelengths through two separate detector channels. These channels are WFC3/UVIS which covers ultraviolet and visible wavelengths and WFC3/IR which covers the infrared wavelengths. The images from WFC3/IR had plate scales of $0.13''/\text{pixel}$, and came from two different filters. Those filters were F110W, a wide YJ filter which covers $9000\text{--}14000\text{ \AA}$, and F160W, an H filter covering $14000\text{--}17000\text{ \AA}$. WFC3/UVIS on the other hand has a plate scale of $0.04''/\text{arcsec}$, both images taken with this instrument used the F475W filter. This is the SDSS g' filter covering $4000\text{--}5450\text{ \AA}$. Data for each galaxy was downloaded from the HST archive¹.

¹<https://hst-docs.stsci.edu/wfc3ihb/appendix-a-wfc3-filter-throughputs>

**Galaxy Sample Properties from NASA/IPAC Extragalactic Database
(NED)**

Galaxy Name	RA (°)	Dec (°)	Classification	Redshift
NGC 1289	50.054515	-2.114078	S0	$0.009313 \pm 1.70\text{e-}5$
NGC 4435	186.918772	13.079024	S0	$0.002638 \pm 1.70\text{e-}5$
NGC 4473	187.453666	13.429419	E	$0.007485 \pm 6.00\text{e-}6$
NGC 4489	187.717702	16.758825	E	$0.003135 \pm 6.41\text{e-}6$
NGC 4552	188.915863	12.556342	E	$0.001134 \pm 1.40\text{e-}5$
NGC 4578	189.377317	9.555069	S0	$0.007645 \pm 1.70\text{e-}5$
NGC 4608	190.305441	10.155719	S0	$0.006171 \pm 1.70\text{e-}5$
NGC 4690	191.981333	-1.656050	S0	$0.009223 \pm 1.70\text{e-}5$

Table 4.1: Information for each galaxy in the sample obtained from the NASA/IPAC Extragalactic Database (NED). Columns include New Galaxy Catalogue name (NGC), right ascension (RA), declination (Dec), morphological classification, and measure of redshift with associated uncertainties.

NGC 1289

NGC 1289 is a lenticular galaxy in the Eridanus constellation². No previous work was done on structural decomposition for this galaxy, but it has been included in surveys including one investigating radial velocities in dwarf galaxies (Heesters et al. (2023)) and another that modeled pure-bulge galaxies (Chae et al. (2019)). Its morphological classification was made by de Vaucouleurs et al. (1991), base on visual observations of the galaxy. This galaxy was outside the range available in SDSS, so no image was present.

²<https://ned.ipac.caltech.edu/>

NGC 4435

NGC 4435 is a barred lenticular galaxy located at a distance of 16.5 Mpc (Tully et al. (2013)) in the Virgo constellation (Binggeli et al. (1985)). It was previously assumed that due to the nearby presence of the disturbed spiral NGC 4438 as well as an arm that appeared to connect both galaxies, that these two galaxies are currently interacting (Vollmer et al. (2005)), but recent work shows that NGC 4435 remains undisturbed. On top of the bar it also hosts a circum-nuclear keplerian disk (CNKD) (Coccato et al. (2004)). A CNKD is a region around the black hole where a higher σ_* can be seen, CNKDs were used as early evidence for the discovery that SMBHs exist at the center of every galaxy (Bertola et al. (1998)). The presence of a CNKD may be indicative that this galaxy requires more than 1 component to be properly fit as the presence of a CNKD introduces at least 2 distinct structures in the galaxy.

NGC 4473

NGC 4473 is an elliptical galaxy in the Virgo cluster. It has been found to have kinematics that behaves differently in the inner and outer regions of the galaxy. This is often an indication that the galaxy may contain a disk (Panetta et al. (2015)). Because of these characteristics it has been said to display characteristics of something between an elliptical and a lenticular galaxy (Bellstedt et al. (2017)). This means that even though the galaxy is bulge dominated it might still display some sort of disk-like structure.

NGC 4489

NGC 4489 is a low luminosity dwarf elliptical galaxy in the Virgo cluster (Jones (1995)). No structural decomposition has previously been performed on this galaxy, but its

classification was performed by de Vaucouleurs et al. (1991) base on visual observations of the galaxy.

NGC 4552

Located in the Virgo Cluster, NGC 4452 is classified as an elliptical galaxy. This galaxy has been previously found to host an AGN (Kara et al. (2024)), specifically classified as a low-ionization nuclear emission line region (LINER) nuclei, a specific type of AGN with very slow variability. This galaxy also has been found to contain a central ring-like structure, with dust streams that flow outwards way past the R_e of the entire galaxy (Bonfini et al. (2018)).

NGC 4578

NGC 4578 is a fast rotating lenticular galaxy (Krajnović et al. (2018)). Prior work found that its kinematics display an elongated structure along the major axis which could be indicative of a bar (Krajnović et al. (2018)).

NGC 4608

At a distance of 17.3 Mpc, NGC 4608 is a massive lenticular galaxy in the Virgo cluster. This galaxy displays a very strong bar (Erwin et al. (2021)). Previous modeling by Erwin et al. (2021) of WFC3 HST Images of the galaxy used 5 components to fit all the structure. They separated it into a bulge, disk, a bar, and a ring. Others found the galaxy to not even contain a disk, meaning it is bulge-dominated similar to an elliptical, and believe the disk to have either been destroyed by the presence of such a strong bar, or for it to have just never existed (Gadotti & de Souza (2003)).

HST Data			
ID	Instrument	PI	Filter
NGC 1289	WFC3/UVIS	Krajinovic	F475W
NGC 4435	WFC3/IR	Boizelle	F160W
NGC 4473	WFC3/IR	O’Robinson	F160W
NGC 4489	WFC3/IR	Blakeslee	F110W
NGC 4552	WFC3/IR	Robinson	F110W
NGC 4578	WFC3/IR	Erwin	F160W
NGC 4608	WFC3/IR	Erwin	F160W
NGC 4690	WFC3/UVIS	Krajinovic	F475W

Table 4.2: Information of the observations used to obtain the data we downloaded for each galaxy from the Barbara A. Mikulski Archive for Space Telescopes (MAST) Portal.

NGC 4690

NGC 4690 is a lenticular galaxy in the Virgo cluster. The only structural component previously identified in literature was both a disk and a ring (Sil’chenko (2016)). This was determined through observations of the inner stellar population properties and gaseous kinematics of the galaxy.

Chapter 5

Galaxy Fitting

For each galaxy in the sample, GALFIT was used to perform a 2-dimensional decomposition following the same standard method. A 2 Sèrsic component fit was attempted, with one trying to account for the bulge ($n = 4$) and the other for the disk ($n = 1$). Other than n the initial value of each parameter for the components would be guessed based on the image of the galaxy. If this fit did not work, the output parameter file and components would be added or removed depending on what seemed necessary. Once the fit either seemed to sufficiently model the light distribution of the galaxy or was not getting any better this last run model would then be used in the analysis for this paper.

Galaxy Fits

Galaxy Name	Component Type	x, y position (pixels)	Magnitude	R_e (arcsec)	Sèrsic Index (n)	Axis Ratio (b/a)	Position Angle	Note
NGC 1289	Sèrsic	2099.91, 2794.21	10.77	5.310	5.16	0.5682	-9.83	Bulge
	Sèrsic	2105.18, 2796.35	10.61	23.061	0.99	0.5651	-6.12	Disk
NGC 4435	Sèrsic	894.40, 820.4142	6.91	5.015	2.0067	0.7010	42.66	Bulge
	Sèrsic	896.42, 821.78	7.84	20.384	0.5174	0.288	51.24	Bar
	Exponential Disk	892.50, 822.82	7.84	36.921	...	0.2880	51.56	Disk
NGC 4473	Sèrsic	598.16, 517.52	8.28	7.141	0.74	0.8987	44.31	Bulge
	Sèrsic	597.33, 519.09	5.2	43.237	5.97	0.4620	-41.67	Disk
NGC 4489	Exponential Disk	592.45, 525.26	8.3	27.77	...	0.6673	-21.57	Pseudo-Bulge
	Sèrsic	595.51, 522.51	6.25	68.733	7.24	0.9508	54.87	Disk
	PSF	594.71, 522.93	10.07	...	...	1.00	0	AGN
NGC 4552	Sèrsic	596.56, 520.48	3.63	76.417	7.01	0.9317	-19.27	Spheroid
NGC 4578	Sèrsic	595.59, 517.73	6.72	20.353	4.92	0.7056	63.46	Spheroid
NGC 4608	Sèrsic	597.34, 523.25	8.11	7.576	0.7766	0.9487	70.64	Pseudo-bulge
	Sèrsic	595.16, 524.73	8.04	29.295	0.4778	0.2816	44.93	Bar
	Sèrsic	596.82, 523.71	7.34	51.823	8.93	0.9118	52.93	Disk
NGC 4690	Sèrsic	2066.63, 2738.81	11.7534	7.842	4.00	0.8034	3.09	Bulge
	Exponential Disk	2064.25, 2730.60	10.47	55.169	...	0.7142	1.60	Disk

Table 5.1: Final fit parameters for each galaxy in the sample

NGC 1289



Figure 5.1: NGC 1289’s decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

The final fit for NGC 1289, shown in Figure 5.1, used 2 Sèrsic components [Table 5.1], which resulted in a $\chi^2_\nu = 0.554$. The first Sèrsic component with $n = 5$ and a $R_e = 5.31$ models the bulge of the galaxy, as classical bulges are often modeled by Sèrsic components with $n > 2$ (Fisher & Drory (2010)). The second Sèrsic which has $n = 0.99$ and is much larger, at an $R_e = 23.06$, models the disk of the galaxy, as disks are often modeled by exponential components, which are just Sèrsic components with $n = 1$.

This two component fit seems to capture the majority of the light from the galaxy. As can be seen in the residual very little obvious structure is left over. The only bright spots that can really be seen are near the center where some light is left over. An effort was made to add components to account for this remaining light, but GALFIT could not converge on a physically reasonable result. We ultimately concluded that, given the minimal residual, this is a physically meaningful fit, and that it account for

the dominant structure in this galaxy.

NGC 4435



Figure 5.2: NGC 4435’s decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

The final model for NGC 4435 [Figure 5.2] required 2 Sèrsic components and an exponential disk component [Table 5.1]. The first Sèrsic component has a Sèrsic index $n = 5$, corresponds to a classical bulge, which are often modeled by $n > 2$. The second Sèrsic component with $n = 0.5$ and axis ratio $= 0.288$ seems to model a bar-like structure. The position angle of this component also matches up with what can be seen in the original image of the galaxy. Finally the exponential component models the disk of the galaxy, as it has the largest R_e of all the components extending out to almost $37''$. The residual image seems to indicate more structure in the disk region of the galaxy, as well as a pattern of overfit and under fit regions towards the center, which could be fixed by the addition of further Sèrsic components, as they are the most general and I could not figure out what this structure was.

This final fit captures the major structure, which includes the bulge, the bar, and the disk. There appears to be some spiral structure towards the center of the galaxy that was not captured by this model. By dealing with that light the pattern at the center of the galaxy may disappear, but while trying to fit it using another Sèrsic component, I was unable to reach a result that properly modeled the other 3 components and removed this light.

NGC 4473



Figure 5.3: NGC 4473's decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

The final decomposition for NGC 4473 [Figure 5.3] used 2 Sèrsic components [Table 5.1]. Most of the literature pointed to a 3 component fit being used to properly model all the components, and I account for the bulge and the disk, but am missing something to account for the light on the top left and bottom right corners of the central region. However, the addition of a third component would always lead to one of the components being sent to unrealistic values, which is why I decided to only use 2 when modeling. The residual for this decomposition shows some over fitting of a

bar-like structure running from bottom left to top right near the center of the image, and the light from this region can be seen in the model image, as there is a bright region running exactly in this direction. Perpendicular to this dark region there is a very bright region which might either be a wide bar or a ring around the center of the galaxy. Having another component to deal with this light might help resolve the over fitting issue of the first bar, and doing so would go along with previous papers which reported this galaxy was well fit by a 3 component model, one described as large and which follows a de Vaucouleurs profile, one fitting the pseudo-bulge at the center, and one orthogonal to the previous 2 components (Lalović & Knežević (2024)). When attempting to fit a third component, GALFIT would send one of the 3 components to a $R_e = 0$, making it not affect the model. This result is why the final fit, contrary to previous research, remains with only 2 components. I believe that this is enough for the purposes of this experiment, even though accounting for this excess light may further reduce the R_e what we have now is sufficient for our analysis.

NGC 4489



Figure 5.4: NGC 4489’s decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

Being one of the AGNs that were able to sneak into this sample of quiescent galaxies, the final fit for NGC 4489 [Figure 5.4] required 3 components to be fit, one exponential, one Sèrsic, and one PSF [Table 5.1]. The exponential component seemed to correspond to a pseudo-bulge as it had a significantly smaller radius than the Sèrsic component. Since exponentials are often used to fit disks, and pseudo-bulges display properties similar to disks it would make sense that this would try to fit that. The Sèrsic component with $n = 8$ may have been fitting the rest of the galaxy as a bulge like structure. This galaxy is an elliptical so it makes sense that it follows a spheroid structure, but this one appears to be split into a more disk-like one towards the center, and a more classical bulge as the remainder of the galaxy. Finally, the PSF component was put there to account for the AGN in the center. Without this component GALFIT would send the Sèrsic index to 20 which creates an extremely sharp profile, which indicated that there was some extremely bright source of light in the center of the galaxy.

This 3 component fit did a fairly good job of accounting for the light in this galaxy. There seems to be some weird pattern of light towards the center of the galaxy which can be seen in the residual, but this may be getting caused by the PSF used in the fit, as it may not be as accurate as a PSF needs to be when fitting an AGN. There also appears to be 1 or 2 concentric ring-like structures that might need require new components to be properly modeled, but overall the majority of the light was extracted making this fit accurate enough for our analysis.

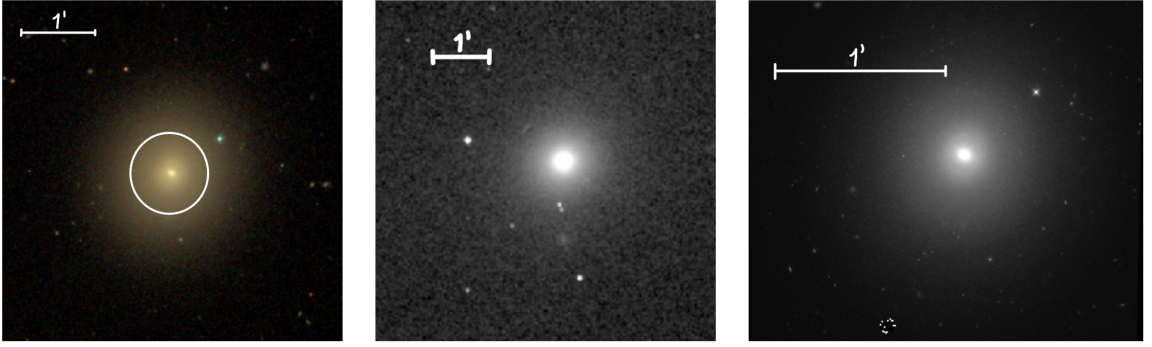


Figure 5.5: SDSS (left), 2MASS (middle), and HST (right) images for NGC 4489. The scale bars show how wide one arc-minute would appear on the image, and the white circle on the SDSS image represents the R_e of the central spheroid.

Even with three components, we found the R_e of this galaxy to be the exact same as that presented by Cappellari et al. (2011). The 2MASS image was what *Atlas*^{3D} used in their analysis and when we look at the image compared to both SDSS and HST in Figure 5.5, we can see that it is unable to capture most of the light in the outter regions of the galaxy, making it appear significantly smaller than what it truly is. This difference is what led to their falsely small radius, and can explain why adding multiple components didn't change this value. The central region is the brightest so it makes sense that out of all the regions this was properly captured in the 2MASS image, but data from HST properly accounts for the remainder of the galaxy.

NGC 4552



Figure 5.6: NGC 4552’s decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

The final decomposition for NGC 4552 [Figure 5.6] required only a single Sèrsic component with $n = 7.01$ [Table 5.1]. When modeling elliptical galaxies we assume that their light profiles are modeled similarly to a bulge component. This cannot always be assumed to be true, as every other elliptical in this sample required multiple components to be fit, but this is one that was fairly accurately modeled by the single Sèrsic. The final χ^2_ν was 4.363, meaning that some of the light from this galaxy was not accounted for in this model. This can be seen in the residual by the ring-like structure surrounding the center of this galaxy, which was confirmed to exist by Bonfini et al. (2018). Any attempts to fit this ring resulted in GALFIT not being able to converge on a physically reasonable result, so we ultimately chose to ignore the ring for the purposes on this paper.

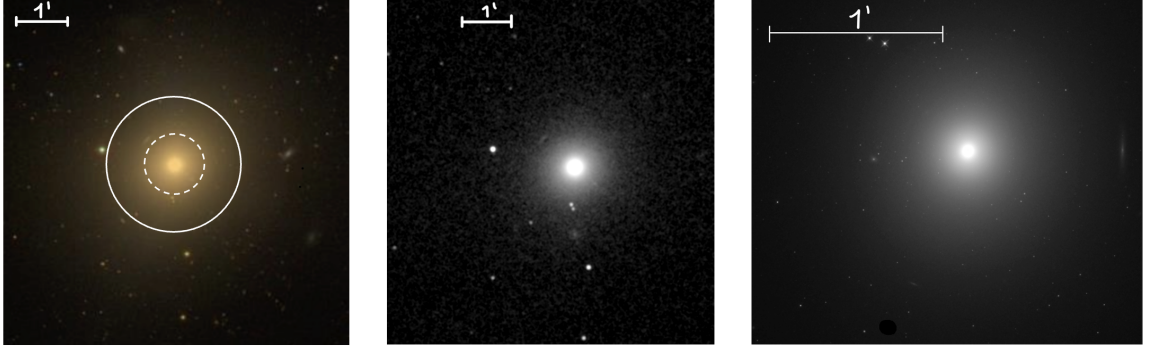


Figure 5.7: SDSS (left), 2MASS (middle), and HST (right) images for NGC 4552. The scale bars show how wide one arc-minute would appear on the image, and the dashed circle on the SDSS image represents the R_e from *Atlas^{3D}*, and the solid white circle is our calculated R_e for the central spheroid.

In Figure 5.7 we are able to compare three images for this galaxy. The 2MASS image, used by Cappellari et al. (2011) does a considerably worse job of capturing all the light from the outer edges of the galaxy than HST, and this can be easily seen when comparing the SDSS and 2MASS images. Because of this when analyzing their galaxy they simply lacked the resolution to capture all of the light, which led to their fit making the galaxy seem significantly smaller than its true size. This can be used to explain why our value for the R_e is almost double what was presented by Cappellari et al. (2011).

NGC 4578



Figure 5.8: NGC 4578’s decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

Even though NGC 4578 is a lenticular galaxy its final decomposition only required 1 Sèrsic component [Figure 5.8]. This Sèrsic component had a Sèrsic index = 6.72 [Table 5.1], which signifies a high concentration of light towards the center of the galaxy. This resulted in a $\chi^2_\nu = 1.147$, which indicates a very accurate fit. Although there might be some structure leftover in the residual which can be seen by the concentric pattern of bright and dark regions, most of the light seems to have been extracted from the original image. This indicates that even though this galaxy should be more complicated, as it has been classified as lenticular, in reality it is very bulge-dominated, and can be accurately modeled by a single component.

NGC 4608



Figure 5.9: NGC 4608’s decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

The final image decomposition of NGC 4608 [Figure 5.9] used 3 Sèrsic components [Table 5.1]. The first Sèrsic component appears to model a pseudo-bulge, as is indicated by its Sèrsic index, $n = 0.7766$. The second Sèrsic appears to model the bar as is indicated by its Sèrsic index being 0.4778 as well as the fact that its axis ratio is 0.2816 which would lead to the component being stretched out, which is exactly what we would expect for a bar, and the position angle for this component is consistent with the orientation of the bar in the galaxy’s original image. The final Sèrsic had the higher R_e and is assumed to have modeled the disk of the galaxy. Its unusually high Sèrsic index of $n = 9$, is a lot different from the usual $n = 1$ that models disks. This is, then, a spheroid rather than a traditional disk. NGC 4608 is one of the most complex galaxies in this sample, and that can be seen in the residual which shows leftover light from almost every structural component present in the galaxy. There seems to be a ring at the end of the bar structure, and there isn’t currently a component accounting for this light. This may explain why the bar and disk components seem to

under fit their specific regions, as they are trying to account for the light coming from the ring on top of the other structural components. Adding a component specifically for this ring may result in all other structures in the galaxy being better fit. This may explain why GALFIT is struggling to fit the entire thing as a large bulge and is instead separating it into multiple spheroids as well as a bar structure.

NGC 4690



Figure 5.10: NGC 4690’s decomposition including the original image (left), the model (middle), and the model subtracted residual image (right). The original image as well as the model are displayed on the same logarithmic scale, and the residual is displayed on a linear scale. The black circle on the original image shows the effective radius of the bulge, as given in table 5.1.

For NGC 4690 a very standard two component fit was used [Table 5.1], resulting in the final fit shown in Figure 5.10. The first component was a Sèrsic with $n = 4$, which corresponds perfectly to a de Vaucouleurs model of a classical bulge. The second component was a much larger exponential disk which corresponds to the disk of the galaxy. This fit resulted in a $\chi^2_\nu = 0.399$ which is fairly low, indicating that the image was overfit, but little evidence of this is found in the residual image. This is one of the

best residuals in this entire sample, as everything appears to be the same brightness, which indicates that the galaxy was properly removed, and the only thing leftover is the sky. Prior experimentation by Sil’chenko (2016) indicated the presence of a ring, but I found no indication of that when trying to model this galaxy.

5.1 Kinematic Plots

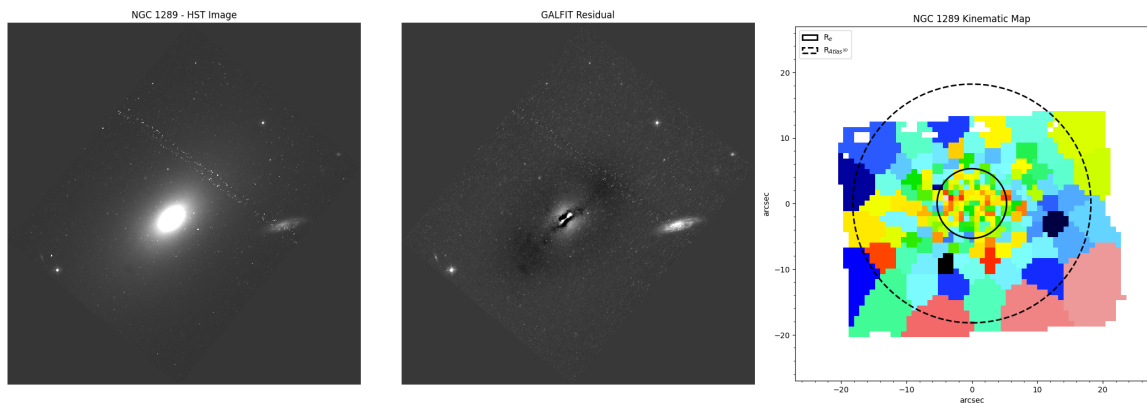


Figure 5.11:

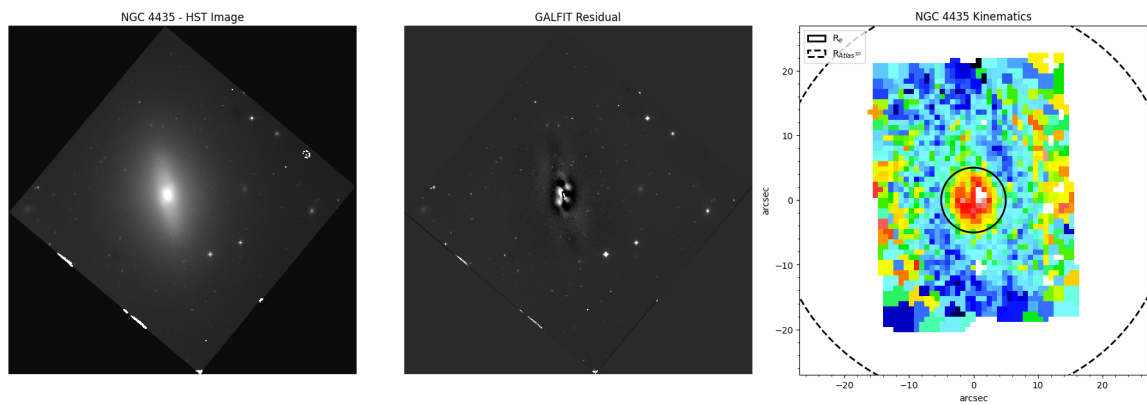


Figure 5.12:

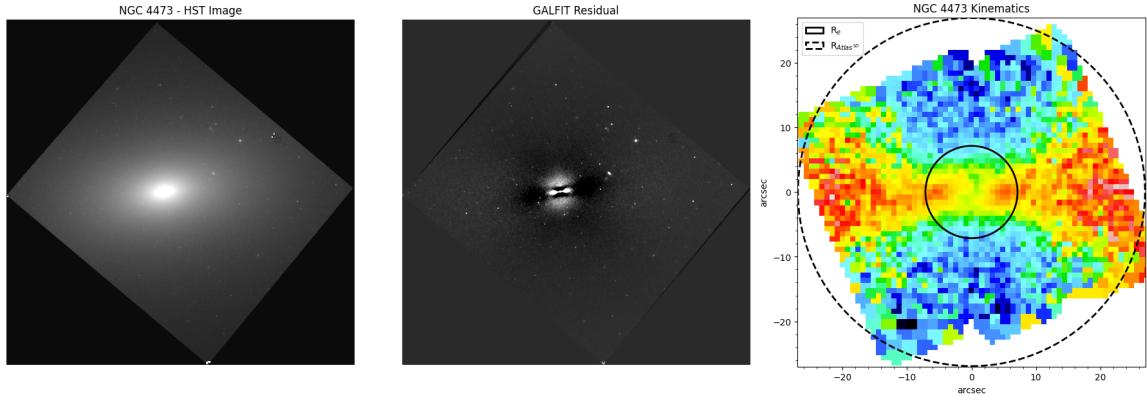


Figure 5.13:

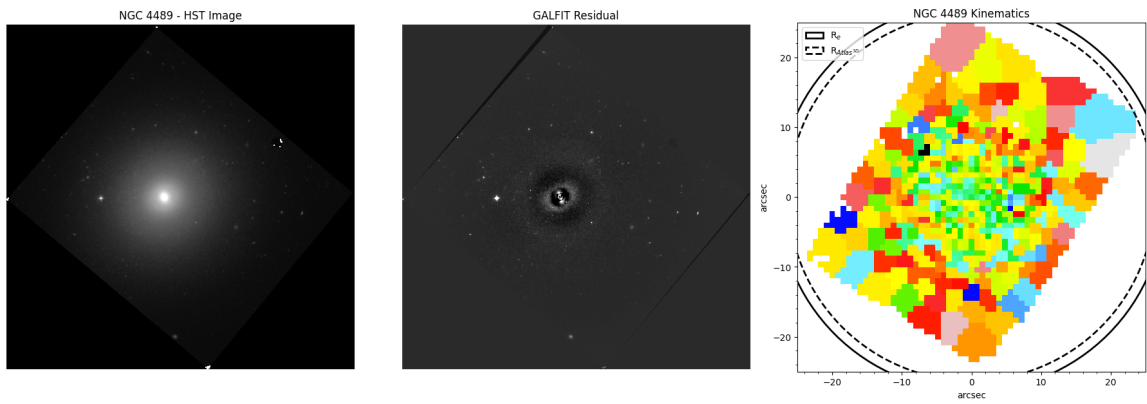


Figure 5.14:

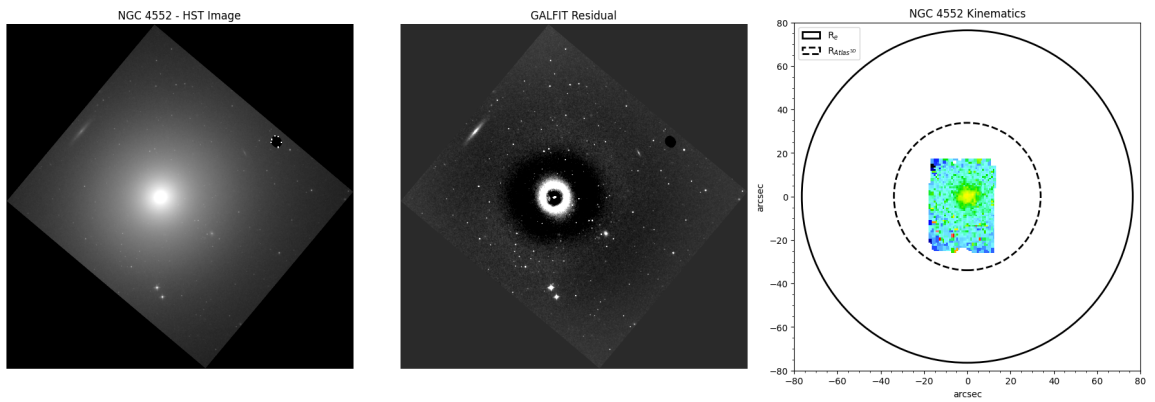


Figure 5.15:

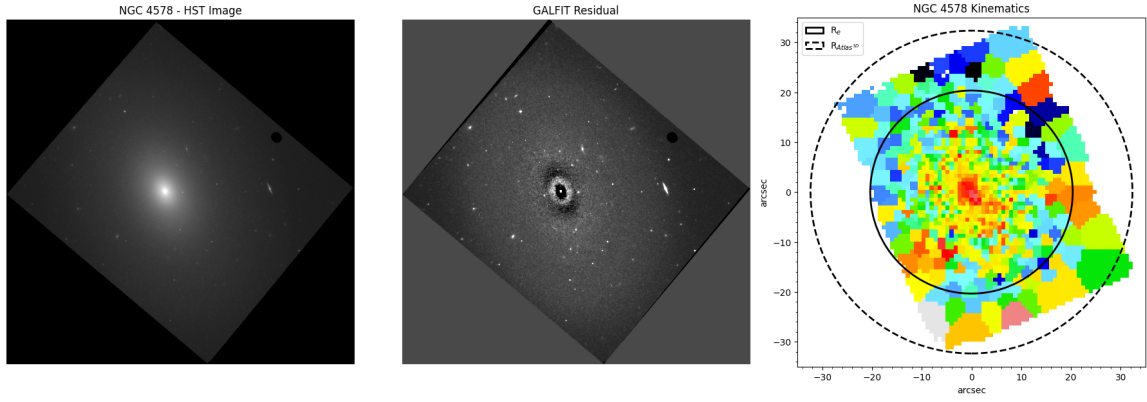


Figure 5.16:

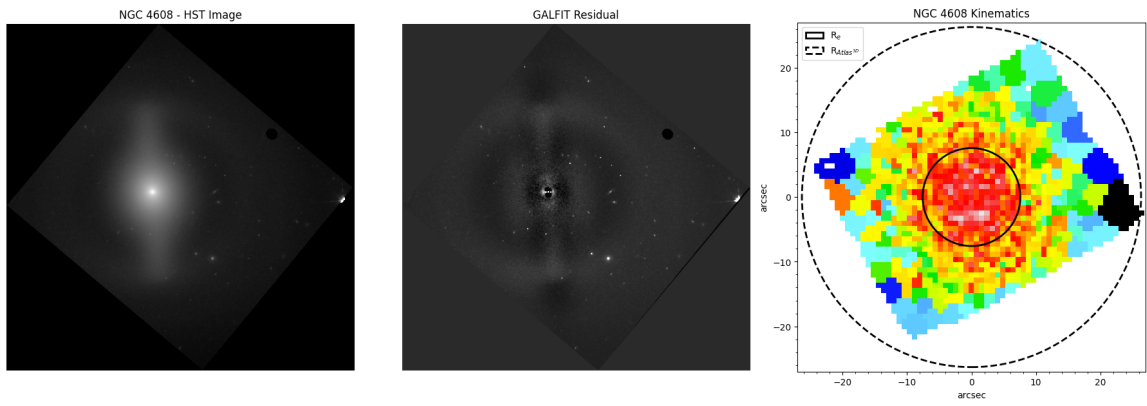


Figure 5.17:

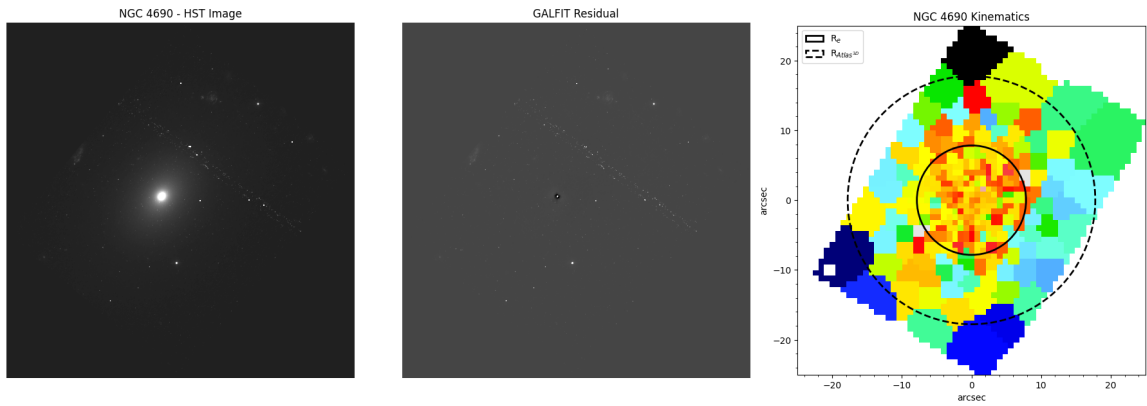


Figure 5.18:

Figure captions: The figures above show the HST image (left), the residual for the best fit model, and the kinematic map (right) for each galaxy. On the kinematic map, both the R_e calculated in this project and in *Atlas*^{3D} are displayed. Each image is aligned so that North points up and East points left.

For each galaxy we made kinematic maps using data downloaded from the *Atlas*^{3D} survey¹. Looking at these kinematics maps we can see that our changing values of R_e can lead to significant changes in the stellar velocity dispersion. By comparing these kinematic maps to the original images we can begin to see how different structures affect the motion of stars in the galaxy and gain a better understanding of what components have the highest effect on the stellar velocity dispersion for a galaxy.

¹<https://www-astro.physics.ox.ac.uk/atlas3d/>

Chapter 6

Analysis

Throughout the multi-component fitting process, we refined measurements of R_e for this sub sample of *Atlas*^{3D} galaxies originally analyzed through single Gaussian fits. Table 6.1 summarizes our R_e measures and compares them to those obtained by Cappellari et al. (2011). By combining the sample of galaxies analyzed in this project with those analyzed by Kesha Patel in her thesis, indicated by an * in the table, we were able to determine that on average a multi-component fit led to a reduction in the R_e by 3.7 ± 0.7 times when compared to the values presented by Cappellari et al. (2011).

Although the average decrease in size was 3.7 ± 0.7 times, this decrease ranged from 1.59 times for NGC 4578, at the lower end of the range, to 10.24 times for NGC 3384, at the upper end. However, most galaxies saw a decrease in R_e of between 2 and 5 times. As mentioned before there were two galaxies that did not follow this general trend. These were NGC 4489 whose R_e remained about the same as the value calculated by Cappellari et al. (2011). The second galaxy was NGC 4552 where instead of the R_e decreasing it actually saw an increase, with the *Atlas*^{3D} value actually being 0.44 times the value calculated in this project. As was described in Chapter 5, these galaxies show the importance of using HST images as opposed to those from ground

Effective Radius (R_e) Values

Galaxy Name	R_e (arcsec)	$R_{e,Atlas^{3D}}$ (arcsec)	$R_{e,Atlas^{3D}}/R_e$
NGC 1289	5.310	18.20	3.43
NGC 4435	5.015	30.90	6.16
NGC 4473	7.141	26.92	3.77
NGC 4489	27.774	26.30	0.95
NGC 4552	76.417	33.88	0.44
NGC 4578	20.353	32.36	1.59
NGC 4608	7.576	26.30	3.47
NGC 4690	7.842	17.78	2.27
NGC 3384*	3.16	32.36	10.24
NGC 3945*	5.96	28.18	4.73
NGC 3998*	5.42	19.95	3.68
NGC 4382*	26.35	66.07	2.51
NGC 4472*	21.29	95.50	4.49

Table 6.1: A comparison of effective radii determined in this project and those determined by the $Atlas^{3D}$ survey. Columns include New Galaxy Catalogue (NGC) name, the R_e determined in this paper, the $R_{e,Atlas^{3D}}$ determined by Cappellari et al. (2011), and finally the ratio between the $R_{e,Atlas^{3D}}$ from Cappellari et al. (2011) and R_e determined in this project. Galaxies followed by an asterisk were obtained from Kesha Patel’s thesis, these galaxies had their effective radii measured by a method similar to this project, differing only in complexity of fit, with hers being more complex than the ones used here.

based telescopes, as even though it results in images with smaller FOV, the amount of light captured is so much greater that even without further complication of the fits, we already see significant changes to the measured R_e .

Even with the two outliers being included in our average we still saw a decrease of the R_e by almost 4 times. Excluding those two galaxies we get an average decrease of 4.2 ± 0.8 , this is a non statistically significant difference from when the two outliers were included. This decrease, was therefore, fairly large, whether or not these outliers were included, and this change came from fairly simple fits. Most of these fits still had significant structure leftover in their residuals, which is indicative of how much more complicated these galaxies are compared to the fits in this study. However simple these fits may be they still demonstrated the importance of properly accounting for galaxy structure in the R_e calculation. Continuing add components would either continue to reduce the R_e even further or not result in a significant decrease.

Even though this sample is small, which makes it difficult to establish any concrete trends, there still appear to be some connections that we can make. It appears that single component fits not accounting for a bar lead to a greater over estimation of the R_e than galaxies without one. This can be seen in both NGC 4435, which was over estimated by 6.16 times in Cappellari et al. (2011), and NGC 4608 which was over estimated by 3.47 times. This is consistent with prior experiments, such as that performed by Gadotti (2008) which showed that not accounting for a galactic bar can lead to a significant over-estimation of the bulge-to-total luminosity ratio for the galaxy, this over-estimation plays a key role in increasing the obtained R_e . On the other end of the spectrum the simpler galaxies such as NGC 4578 which was fit by a single Sersic, and NGC 4690 which was very accurately fit by a 2 component fit, had smaller decreases in R_e compared to the average, those decreases being 1.59 and 2.27 times respectively.

The overestimation for simpler galaxies is less than that for complicated galaxies,

but the presence of this effect in even the simplest of galaxies shows the importance of properly fitting galaxy structure when attempting to constrain values of σ_* . Due to this fact, I believe that taking the time to develop more complicated models for this quiescent sample would lead to a significant increase in the accuracy of σ_* measurement, and therefore would allow for a better constrained $M_{BH} - \sigma_*$ relation to be found.

Chapter 7

Conclusion

In this project, we aimed to increase the accuracy of effective radius (R_e) measures for a sample of early-type galaxies through multi component photometric decomposition using the software GALFIT. The galaxies investigated were taken from the Cappellari et al. (2011) sample, that was previously studied, and shown to have inaccurate R_e measures by Batista et al. (2017b). Our analysis showed that these multi component fits consistently yielded smaller R_e measures, on average by a factor of 3.7 ± 0.7 times smaller than those reported by *Atlas*^{3D}. Even though these fits were relatively simple compared to what can be accomplished with more effort, I believe that the single Gaussian fits previously performed lead to a systematic increase in the measured R_e for this sample of galaxies which led to significant inaccuracies when using those values to obtain σ_* values for these galaxies.

From these simple fits we also showed that the application of a correction to the *Atlas*^{3D} values would not be sufficient to properly constrain the values, as even in this small sample the change in R_e ranged from 0.4 times to 10 times. From this result, we determine that each galaxy to be used in the calculation of σ_* would require its own complex fit.

Finally, we saw the importance of using HST to obtain the images of the galaxies

being analyzed. The significantly higher resolution of HST when compared to previously used ground telescopes, such as 2MASS, allow much more of the light from the galaxy to be captured. This reduces any error that would arise from a galaxy's image missing the majority of the light from the outer and dimmer regions.

Future work would be: to calculate a new measure of σ_* for the galaxies analyzed in this sample, further refine the fits for this sample of 8 galaxies, expand the work done in this paper to the 32 quiescent galaxies analyzed by Batiste et al. (2017b), and finally apply these more complex fits to images from 2MASS to determine whether the fits or simply the improved images lead to our corrected values of R_e .

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